Divergent mean heights on a nonpreperiodic quadratic orbit

Submission for TLMC problem 00000002689

24 October 2026

Scope. The conjecture asserts that the mean naive height of an orbit outside the preperiodic set converges, with a proposed canonical-height expression and rate. We use the ordinary arithmetic mean over the first n iterates, without degree normalization. For the degree-two polynomial $f(z) = z^2$ over $\mathbb{Q}$, the orbit of 2 disproves this necessary convergence assertion. Both standard conventions for naive height, logarithmic and multiplicative, give means tending to $+\infty$. We do not specify the source's undefined Fourier coefficients or orbit-frequency measure, and make no claim against a different, degree-normalized averaging statement.

Actual heights and iterates. For a rational number a/b in lowest terms with $b > 0$, the standard multiplicative and logarithmic Weil heights are

$$H(a/b) = \max\{|a|, b\}, \qquad h(a/b) = \log H(a/b).$$

The formal proof uses Mathlib's native heights defined from the archimedean and nonarchimedean absolute values. The displayed rational formulas are proved consequences of those definitions [1]. In particular, no orbit height is assigned by definition.

Take $f(z) = z^2$ and $x_0 = 2$. The actual iterates satisfy

$$x_k = f^k(2) = 2^{2^k}, \qquad H(x_k) = 2^{2^k}, \qquad h(x_k) = 2^k \log 2.$$

The sequence x_k is strictly increasing. Its forward orbit is therefore infinite, so 2 is not preperiodic. The formalization constructs the actual polynomial X^2, proves its degree is two and its evaluation is f, and defines preperiodicity by finiteness of the actual forward orbit.

The canonical height is finite. The canonical height of a degree-$d > 1$ map is obtained from the normalized height limit [2]. Here, for every $x \in \mathbb{Q}$ and every $k \in \mathbb{N}$,

$$\frac{h(f^k(x))}{2^k} = h(x).$$

Consequently the genuine normalized limit exists and

$$\widehat{h}_f(x) = \lim_{k\to\infty} \frac{h(f^k(x))}{2^k} = h(x), \qquad \widehat{h}_f(2) = \log 2 > 0.$$

In Lean the limit is defined by `Filter.limUnder`; convergence is proved before its value is identified. Thus the finite canonical height comes from its defining limit, not from a numerical placeholder.

The unnormalized mean diverges. For $n \geq 1$, let

$$A_n = \frac{1}{n} \sum_{k=0}^{n-1} h(x_k).$$

The actual finite geometric sum gives

$$A_n = \frac{2^n - 1}{n} \log 2.$$

For completeness, an elementary lower bound suffices to establish divergence without an asymptotic approximation. The inequality $2^k \geq k + 1$ implies

$$A_n \geq \frac{\log 2}{n} \sum_{k=0}^{n-1} (k + 1) = \frac{n + 1}{2} \log 2 \longrightarrow +\infty.$$

Therefore A_n cannot converge to any finite real number. In particular, it cannot have the finite limit asserted in the conjecture, and hence cannot converge to such a limit at rate $1/\log n$.

If "naive height" instead means H, the conclusion is unchanged. Because $h(x) = \log H(x) \leq H(x)$ and $H(x) \geq 1$, the arithmetic means

$$B_n = \frac{1}{n} \sum_{k=0}^{n-1} H(x_k)$$

satisfy $B_n \geq A_n$ and also tend to $+\infty$. The formal conclusion excludes convergence to every real number for both sequences.

Formal scope and verification. The complete proof is in `lean/Main.lean`, culminating in `TLMC2689.counterexample`. The statement connects the actual degree-two polynomial, all its iterates, the nonpreperiodic orbit, the native heights, the canonical-height limit and both divergent means. All convergence assertions concern $n \to \infty$; the lower-bound argument uses $n \geq 1$ throughout. No assertion depends on a zero-denominator value.

The original bilingual conjecture is preserved in `SOURCE.md`. The project pins Lean 4.33.0 and Mathlib commit

`db584cd6d46c92f209a44c0f1c829460d327499d`.

The accompanying evidence records the exact build and exhaustive namespace axiom audit, a fresh kernel replay, rendered-PDF review and file hashes. The disproof concerns the ordinary, unnormalized mean only; it does not dispute canonical-height theory or arithmetic equidistribution under their hypotheses.

References

[1] Mathlib, *Heights on number fields*, pinned source, rational-height formulas.

[2] L. DeMarco, *Arithmetic Dynamics and Intersection Problems*, Arizona Winter School 2023, §1.4, p. 6, author's lecture notes.